

- 10** The plane  $P$  has normal vector  $2\mathbf{i} + a\mathbf{j} - \mathbf{k}$ , where  $a$  is a positive constant, and the point  $(3, -1, 1)$  lies in  $P$ . The plane  $x - z = 3$  makes an angle of  $45^\circ$  with  $P$ .

Find the cartesian equation of  $P$ .

**[7]**